

UNSUCCESSFUL & DRY OIL/GAS WELLS

Documented cases from Sivasagar, Assam and Mehsana, Gujarat

Project Report

Purpose: A simplified explanation of selected documented drilling failures, dry wells, and abandoned exploration sites.

1. Introduction

Oil and natural-gas exploration is uncertain. A company may spend substantial money drilling a well and still find no commercially useful hydrocarbons. Such a well is commonly described as a **dry well**. A well may also be abandoned because drilling becomes technically difficult or because the expected reservoir cannot be developed economically. Therefore, “abandoned” does not always mean “no oil was ever present.”

This report focuses on documented examples associated with the Sivasagar/Upper Assam petroleum region and the Mehsana area of Gujarat. The examples are kept separate because a dry well is an individual well result, while an entire oil field can contain both successful and unsuccessful wells.

2. Sivasagar, Assam

Sivasagar is an important petroleum-producing area of Assam. Historical petroleum activity in the district includes areas such as Geleki, Amguri and Lakwa. Scientific literature has also documented abandoned crude-oil-contaminated drill sites at these locations. However, an abandoned drill site should not automatically be called a dry well: the reason for abandonment can differ from well to well.

2.1 East Lakhibari-7 — reported dry well

ONGC reported **East Lakhibari-7** as a dry well during its FY 2025–26 exploration work in the Assam Shelf. This is a useful recent example of exploration uncertainty: another nearby exploration well, East Lakhibari-6, was reported to have discovered hydrocarbons, while East Lakhibari-7 was dry.

Simple meaning: The geological area had petroleum potential, but this particular well did not establish a useful hydrocarbon accumulation.

2.2 GK-11 — technical drilling failure/dry well

The C&AG; of India reported that ONGC planned to explore the Barail/Kopili strata at **GK-11 in Upper Assam**, with a target depth of about **4,710 m**. Drilling started directionally in June 2002. The operation encountered serious **torque and drag** problems. By September 2004 it was not feasible to continue beyond about **4,230 m**, and the rig was released on 7 December 2004. The well did not achieve its exploratory objective and ONGC charged about **■26.02 crore** as expenditure on a dry well.

In simple words: The drill was supposed to go deeper, but twisting and friction forces became too high. Drilling could not safely/practically continue to the planned depth, so the exploration objective was not achieved.

Important location note: The C&AG; document identifies GK-11 as being in Upper Assam, but the accessible report text does not provide enough information to prove that GK-11 was specifically inside Sivasagar district. It is therefore listed here as an Upper Assam case, not as a confirmed Sivasagar-only well.

2.3 Abandoned drill sites: Geleki, Amguri and Lakwa

A scientific study documented abandoned crude-oil-contaminated drill sites at **Geleki, Amguri and Lakwa** in Sivasagar district. These sites demonstrate the long history of drilling and petroleum operations in the district. The word “abandoned” alone, however, is not enough to determine whether each original well was dry, technically unsuccessful, depleted, or uneconomic.

3. Mehsana, Gujarat

Mehsana lies in the petroleum-rich Cambay Basin region. Several fields around Mehsana have had successful production as well as unsuccessful individual wells. The following cases are directly documented by field/operator information.

3.1 Allora Field — three dry wells

The **Allora Field** is about 8 km northwest of Mehsana town. The field was discovered in January 1967 by ONGC through exploratory well **Al #1**. Three additional ONGC wells were drilled: **Mehsana-7, Mehsana-16 and Dhijoj-4**. All three were found **dry and abandoned**.

Simple sequence: Al #1 discovered the field → additional wells were drilled → Mehsana-7, Mehsana-16 and Dhijoj-4 were dry → those three wells were abandoned.

The field itself was not a failure. Al #1 was later reactivated, and after hydraulic fracturing in 2013 it reportedly reached a peak oil production of about 60 barrels per day. This shows why an unsuccessful individual well should not be confused with failure of the whole field.

3.2 Unawa Field — Un #8

The **Unawa Field** is about 20 km north of Mehsana town. ONGC drilled two wells: **Un #4** and **Un #8**. Un #4 proved to be an oil producer from the Olpad Formation, while **Un #8 was found dry and abandoned**.

Simple meaning: Two wells were drilled in the same field, but one found producible oil and the other did not.

3.3 North Balol Field — NB-6 and NB-9

The **North Balol Field** is about 16 km west of Mehsana town. The field has nine wells: NB-1 to NB-8 were drilled by ONGC, and NB-9 was drilled by the later JV consortium in 2007. **NB-6 and NB-9 are documented as dry wells**. Other wells in the field had gas production or produced thick oil during testing.

Simple meaning: North Balol was not completely unsuccessful; it had productive wells, but NB-6 and NB-9 did not provide useful hydrocarbon production.

3.4 Sobhasan Field — four dry wells

A World Bank project document reported that **58 wells** had been drilled in the Sobhasan Field: 42 oil wells, 2 gas wells, 5 observation wells, **4 dry wells**, and 5 wells awaiting further production testing. The field itself was discovered and brought into production in 1969.

Simple meaning: Sobhasan was a successful producing field overall, but four of its drilled wells were classified as dry.

4. Summary Table

Location	Field / Well	Result	Main point
Sivasagar, Assam	Geleki / Amguri / Lakwa	Abandoned sites	Historical drill sites; abandonment does not by itself prove a dry well
Upper Assam	GK-11	Dry / objective not achieved	Directional drilling faced high torque and drag; stopped before target depth
Assam Shelf	East Lakhibari-7	Dry	ONGC reported the well dry in FY 2025–26
Mehsana, Gujarat	Allora: Mehsana-7	Dry & abandoned	One of three additional ONGC wells in Allora
Mehsana, Gujarat	Allora: Mehsana-16	Dry & abandoned	One of three additional ONGC wells in Allora
Mehsana, Gujarat	Allora: Dhijoj-4	Dry & abandoned	One of three additional ONGC wells in Allora
Mehsana, Gujarat	Unawa: Un #8	Dry & abandoned	Un #4 was an oil producer
Mehsana, Gujarat	North Balol: NB-6	Dry	Field also contains producing wells
Mehsana, Gujarat	North Balol: NB-9	Dry	Drilled by JV consortium in 2007
Mehsana, Gujarat	Sobhasan	4 dry wells	58 wells reported; field overall was productive

5. Why Do Oil Wells Fail?

1. **No oil or gas at the target:** Geological predictions are not always correct.
2. **Poor reservoir rock:** Oil may be present, but the rock may not allow enough fluid to flow.
3. **Unexpected water:** A well can encounter water rather than commercially useful hydrocarbons.
4. **Drilling problems:** High torque, drag, stuck equipment, unstable formations, pressure and temperature can make a well difficult to drill.
5. **Economics:** A well can contain hydrocarbons but still be unsuitable for commercial production if recovery costs are too high.

6. Key Takeaways

- A **dry well** is an unsuccessful exploration result for that particular well; it does not mean the entire surrounding field has no petroleum.
- Mehsana has several clearly documented dry wells, including Mehsana-7, Mehsana-16, Dhijoj-4, Un #8, NB-6 and NB-9.
- In Upper Assam, GK-11 is a well-documented example where drilling complications prevented the exploratory objective from being achieved.
- Sivasagar has historical abandoned drill sites at Geleki, Amguri and Lakwa, but “abandoned” should not automatically be interpreted as “dry.”
- Exploration involves uncertainty, so successful and unsuccessful wells can occur within the same petroleum region.

7. Sources

1. Comptroller and Auditor General of India, Audit Report No. 11 of 2007 — drilling of GK-11 in Upper Assam. The report records the target depth, drilling period, torque/drag problems, release of the rig, and dry-well expenditure.
2. Gujarat Natural Resources Limited (GNRL), Allora Field — field history and identification of Mehsana-7, Mehsana-16 and Dhijoj-4 as dry and abandoned.
3. GNRL, Unawa Field — Un #8 documented as dry and abandoned; Un #4 documented as an oil producer.
4. GNRL, North Balol Field — NB-6 and NB-9 documented as dry wells.
5. ONGC-related FY 2025–26 reporting — East Lakhbari-7 reported as dry during Assam Shelf exploration.

Note: This report intentionally distinguishes confirmed dry wells from merely abandoned drill sites. Publicly available sources do not establish a complete list of every unsuccessful well ever drilled in these districts.